Dream, I've been working on Dream challenges for the past 15 years. I'm the vice chair of Dream and working with Gustavo and Julio for a long time. So it's a pleasure to be here and to see so many people. I'm going to talk briefly about the AI Alliance and how we are trying to use it for benchmarking foundation models on drug discovery. So what is the AI Alliance? Well, you know, it's a consortium of companies and universities and different types of nonprofits who really came up two years ago when the foundational models exploded. And the idea is they have key initiatives. This goes well beyond biology or chemistry or medical science. But the idea is that data needs to be, there need to be defined requirements and building an open, trusted data catalog for training and tuning. So everything is transparent and open. At DREAM, I've been working on DREAM challenges for the past 15 years. I'm the vice chair of DREAM and working with Gustavo and Julio for a long time. So it's a pleasure to be here and to see so many people. I'm going to talk briefly about the AI Alliance and how we are trying to use it for benchmarking foundation models on drug discovery. So what is the AI Alliance? Well, you know, it's a consortium of companies and universities and different types of nonprofits who really came up two years ago when the foundational models exploded. And the idea is they have key initiatives. This goes well beyond biology or chemistry or medical science. But the idea is that data needs to be, there need to be defined requirements and building an open, trusted data catalog for training and tuning so everything is transparent and open. There are benchmarks where you can test, evaluate, and monitor foundational models to ensure interoperability, safety, and fairness. And also there's domain expertise to leverage deep domain knowledge and skills into multimodal, domain-specific models and applications. So I guess that sounds very much to things that we are in favor of and are necessary for understanding science and being able to share data and models. Now, when you form a new group, you know, you have to make it happen around something. And we know that very well in DREAM. We're actually surprised that every time we organize a challenge, people come participate. And we've been doing that for 15 years. So one of the reasons is because you need a very focused question that people can participate. Not only that, you need a collaboration and you need results, right? So in order to make the AI Alliance something workable, we decided to organize a subgroup, which is a group that has to do with drug discovery. And in order to organize a subgroup, you need a meeting, just like here. And so last year we organized a meeting where about 160 people came to Boston where we were discussing things that are actually pretty similar to what we're discussing today around foundation models and drug discovery. But really what you need are models and problems to solve them. And so here are just four examples of models that were recently made public with the code, open sourced, and with the papers. All these models are from IBM, from our groups, focused on the four topics that we think are very important for drug discovery. I'll talk briefly about the models. But basically, MELEN is focused on molecules. Mammal can do multiple tasks. It's a T5 architecture, so it can do both molecules but also gene expression prediction. BFMRNA is a specific single-cell transcriptional model, and BFM DNA, which focuses on

DNA a little bit how of the things that Ancho was describing. So the models have kind of different architectures and try to cover both the space of molecules and the space of biology. And really, one of the questions that I wanted to try to ask is the question that Gustavo posed is, how do we advance a benchmarking agenda and what is the benchmarking agenda? And so in a way, I think that we can summarize that in three things. First of all, well, you have to share models, make models available in a certain way that they're usable, and I'll give an example of that. The second is you have to use the models. Ourselves, the people that develop the models, have to show that their models work in specific conditions. Of course, it's great when people use the models, like Anshu was showing, using the foundational models for specific tasks, where in general they seem to be underperforming, depending on how much data you fine-tune, et cetera. And also, it's very nice when, like what Maria was showing, the metric is also very important. So both things, how you implement the model, how you use it, and what the metric is. And third, I think that's also very important, and I think it was also discussed today, is how to generate new benchmarks. The important thing is not only the, if you want, general benchmarks that exist for foundation models, but specific ones that are the ones that are actually important and where you may actually really see the difference or the lack of advantage of foundation models. So these are the three points I want to briefly illustrate with the models that we have developed. So share models. I think it's really important to make models available so that people can use. And Mellon comes a little bit with a philosophy that we have followed in DREAM, which is, you know, it uses molecular images, smiles, so basically just text, and molecular graphs. And it's an ensemble model that has the three, uses the three types of data to generate an ensemble model. You know, the pre-training of the image is, and it's really, you know, the first time I heard about this, it's really an image of a molecule, and you use that as a, you know, with a CNN architecture. Of course, for the SMILES, it's just a text base, and graph is a graph pre-training. And then this is, so you make an ensemble model, and actually, one of the things that we've seen lately in DREAM is that ensemble models are more and more difficult to show a real performance above the best performing models. And really, I think what's happening now in DREAM is we're really going deeper into trying to mix and match different parts of the models to make the models better. And I think a very good example is one of the latest publication in Nature Biotech of this gene expression-based DREAM challenge that we did, and where Rafi, who's here from the DeBerse group, really kind of took the different parts of the architectures to make an ensemble model that's more than just an average of predictions, but really kind of putting the different parts together. So it was a huge effort from him. This is a more classic version of that. And then MAMML is the other model that, as I said, it's a text-to-text. It's interesting because you can basically prompt the model and make it do different things through their encoder-decoder architecture. So you can ask for binding affinity, which organism and which molecular entities. So you basically can predict, and you can benchmark it on the very similar benchmarks, chemical benchmarks and biological

benchmarks that the Mellon model was done with NAML. But really, as I said, what we want is where can you use these models? And so what we did, together with the working group of the Alliance and DREAM, is let's give the models in an interesting problem. And maybe you're aware that we're almost finishing this challenge that we started doing with Target35 and the SGC. The challenge is a relatively simple challenge, but with an amazing amount of data. You have a 300K molecule Dell library where we don't share the smiles in this case. We only share the fingerprints. And you have to predict an orthogonal set of molecules, whether they bind to this WDR91 protein. The challenge went into different steps. The first one is only using fingerprints. The second step was we only gave SMILE for the test set. So the idea is you could do maybe some docking. Those two parts of the challenge are done. And surprisingly, the docking, giving the smiles didn't really help. And so for this challenge, we gave the link to the participants for the models, and we asked them to see if they could use it. So I think that's the first idea I wanted to share is organize a challenge where foundational models are given to participants. And we know in Dream that if you give a model to someone, especially students and people that are very active and want to develop that, they do use it. So I think that's the first step. Have an interesting problem and suggest some models to be used. The other is implement our own models. And of course, I cannot implement my models in Dream Challenges because I can't wear two hats. That becomes a white box that's not the type of white boxes that we want. And so what we're doing is we want, right now our team at IBM is implementing our BMF mRNA transcription model. I'm sure everybody knows about this virtual cell challenge, which Anchil rightly pointed, why is it a virtual cell? It's just a perturbation, right? You just have a set of retrieves, but it's really a great challenge. We're actually jealous because we've been thinking of organizing something like that. I think there's going to be a lot of space for these types of challenges. But just on my second example, our team is participating in that, and this is also a way of showing how your models work. Finally, the third is generate new benchmarks. And this has to do with the animation left, but it doesn't matter. This is a model that we just published last week, and I just shared the links today because now the code is working, everything is working, and I would like people to use it. The idea is we did something a little bit different. We got inspired by SNP2Vac. And what we wanted is to train a DNA-based model that used just masked language, the same thing, but that inherently knows where SNPs are. And so SNPs, whenever there's a SNP, we change it with a Chinese character. We also include deletions and insertions because we think they're important. And so we train the model to try to recognize the SNPs represented as Chinese characters. So in a way, it's a DNA-based model that not only has ATCNGs or tokens representing a combination of the TCNGs, but the tokens also have Chinese characters that represent the SNPs. And if you can see in the benchmarks up there, we're close to DNA BERT on all these different fine-tuned tasks. We actually added a couple. We added the NPRA predictions, and we also added a SNP-to-disease, which we think is a very interesting task. So you see, we're close, but what's really interesting, if you see the two last

columns, is the version of our model that only uses a reference model is worse than the version that uses the SNP model, right, for all the tasks, basically. And I think that's a very important point. Now, in all the things that I'm showing you there, the input is only DNA, is not DNA with Chinese character. And so my last point is we need help for finding, transforming the reference sequences into sequences that have Chinese characters. And I don't know why it's not appearing, but basically imagine a sequence at the same but with Chinese characters. And we also need help for trying to understand what the correct negative class, for example, a promoter classification is. In this case, what they did is they take the promoter, they take chunks of 12 base pairs, they randomize the 12 base pairs, and they say that's the negative class for the promoter. Okay, our problem is, I just missed all my slides, I don't know what happened. Okay, our problem is how do you impute SNPs in that negative class? Do you just impute random SNPs? Do you impute the SNPs that align exactly to the places where there's only negative? Or what else can you do? And so what we show, and I don't know what happened with my slide, but basically if you input SNPs randomly, your model does better. So in a way, it is seeing the differences. And the most, I think, shocking example is we created a new negative task, which is just a positive promoter where you randomize SNPs. And when you do that, your model does much better if you pre-train it than with a model that's not pre-trained. So in a sense, it's really learning when the SNPs are. And we think that there's more applications for that. And what we want is for you to suggest, to take the code and suggest what other fine-tuning applications could be interesting. Okay, I'm done. This is the acknowledgement slide. My life is divided between IMEM and Dream, but we want to make them together. And maybe the AI alliance will help for that. Thank you if you have any questions. Thanks, Pablo. So I will also ask questions in the online system, but we don't have a chance to ask personal questions before we answer the later. I think better than that. Thank you. Hello, the problem is we don't. I can read it to you. Any drugs discovered by your methods or other AI methods? We're finishing the challenge, the target 35 challenge, and the results are going to be actually measured experimentally. So I'll have a response for that in a couple of months. Thank you. Our benchmarks where you can test, evaluate, and monitor foundational models to ensure interoperability, safety, and fairness. And also, there's domain expertise to leverage deep domain knowledge and skills into multimodal, domain-specific models and applications. So I guess that sounds very much to things that we are in favor of and are necessary for understanding science and being able to share data and models. Now, when you form a new group, you have to make it happen around something. And we know that very well in DREAM. We're actually surprised that every time we organize a challenge, people come participate. And we've been doing that for 15 years. So one of the reasons is because you need a very focused question that people can participate. Not only that, you need a collaboration and you need results, right? So in order to make the AI Alliance something workable, we decided to organize a subgroup, which is a group that has to do with drug discovery. And in order to organize a subgroup, you need a meeting,

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